

Skills Summary

Signs and Denominators in Conic Equations

Use this reference while completing your worksheet or when studying.

Skill 1 — The sign flip in the centre

Standard form subtracts: $(x - h)^2 + (y - k)^2 = r^2$. So a plus inside a bracket means a negative coordinate — $(x + 5)^2$ gives $h = -5$.

From the expanded form $x^2 + y^2 + px + qy + f = 0$, each centre coordinate is half its linear coefficient with the sign reversed: $h = -p/2$, $k = -q/2$. Two moves, halve and reverse, and a quoted centre that matches the linear coefficients has skipped both.

And the radius is a square root: $r^2 = (p/2)^2 + (q/2)^2 - f$.

Example: Find the centre and radius of $x^2 + y^2 + 10x - 4y + 13 = 0$.

Step 1. Nothing can be read off until both squares are completed, so halve each linear coefficient, put it inside a bracket, and repay what that adds.

Step 2. Half of 10 is 5 (adding 25); half of -4 is -2 (adding 4): $(x + 5)^2 + (y - 2)^2 = -13 + 25 + 4$.

Step 3. Reading against $(x - h)^2 + (y - k)^2 = r^2$: the brackets flip the printed signs, and the right-hand side is 16.

$\Rightarrow$ centre $(-5, 2)$, radius 4

Try it: Find the centre and radius of $x^2 + y^2 - 8x + 2y - 8 = 0$.

check: centre $(4, -1)$, radius 5

Skill 2 — Which denominator is which

Compare, do not assume. In $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ the *larger* denominator carries the major axis — it is not always the one under x^2 . If the larger sits under y^2 , the major axis is vertical.

Every denominator is a square, so take a square root before reporting a length. The same goes for the right-hand side of a circle equation: it is r^2 , never r .

Example: Find both semi-axes of $\frac{x^2}{9} + \frac{y^2}{25} = 1$.

Step 1. The first question is which denominator is larger, because that decides which axis is the major one — reading left to right would answer it wrongly here.

Step 2. $25 > 9$, so the major axis is vertical with half-length $\sqrt{25}$, and the minor half-length is $\sqrt{9}$.

$\Rightarrow$ semi-major 5 (vertical), semi-minor 3

Try it: Find both semi-axes of $\frac{x^2}{49} + \frac{y^2}{16} = 1$, and say which way the major axis runs.

check: semi-major 7 (horizontal), semi-minor 4

Skill 3 — Add or subtract for the foci

Ellipse subtracts: $c^2 = a^2 - b^2$, and the foci end up *inside* the curve, so c must come out smaller than a .

Hyperbola adds: $c^2 = a^2 + b^2$, and the foci end up *beyond* the vertices, so c must come out larger than a .

Use that as your check. If an ellipse's focus lands outside its vertex, or a hyperbola's lands inside, the wrong rule was used — no further arithmetic needed.

Example: Find the focal distance of $\frac{x^2}{25} + \frac{y^2}{9} = 1$, then of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

Step 1. The operation between the two terms is the only thing that decides add or subtract, so read the sign before writing anything down.

Step 2. Plus sign, so an ellipse: $c = \sqrt{25 - 9} = \sqrt{16}$, which is less than the vertex at 5.

Step 3. Minus sign, so a hyperbola: $c = \sqrt{16 + 9} = \sqrt{25}$, which is more than the vertex at 4.

$\Rightarrow$ ellipse $c = 4$, hyperbola $c = 5$

Try it: Find the focal distance of $\frac{x^2}{100} + \frac{y^2}{36} = 1$, then of $\frac{x^2}{64} - \frac{y^2}{36} = 1$.

check: 8 and 10